

Introduction to Product Management & the PM Ecosystem

Session 01 · Class 0 · Prof. Saravana Jaikumar · 13 Jul 2026

AI has changed PM in three independent places — your own work, your product’s knowledge, and how the product gets built. One framework holds the whole programme together.

AT A GLANCE

- The **Three-Layer Framework** maps every future class: Productivity Amplifier, Product Intelligence, Full-Stack Product Creation.
- The technical shift underneath: **task-based models** → **foundation models**. One pre-trained model replaces building one model per task.
- An LLM does not *know* answers. It **predicts the next likely word**. Context shifts the probabilities — that is why prompting works.
- **Firm ≠ model ≠ platform**. OpenAI is a firm. GPT 5.6 is a model. ChatGPT is a platform. An interview collapsed over this exact confusion.
- Task-based models are **not obsolete** — they are smaller and cheaper. Knowing when a small model beats an LLM call is a real PM decision.
- Your product’s **knowledge base was your moat, and the moat is gone**. The differentiator moves to how deeply you understand a specific user’s process.
- The PM’s definition: **“Here’s the problem. Here’s what we’re building. Here’s how we’ll know it works.”**

THE THREE-LAYER FRAMEWORK

Layer	Covers	AI replaces
1 · Productivity Amplifier	Your own daily PM work — docs, reports, triage	Manual coordination and grunt work
2 · Product Intelligence	Knowledge inside your product	8–12 months of hand-curated data
3 · Full-Stack Creation	How the product gets designed, built, shipped	Long build cycles across specialist teams

The layers are **independent** — there is no architecture to wire between them. In every class, name which layer you are sitting in.

TASK-BASED VS FOUNDATION MODELS

	Task-Based	Foundation
Scope	One task only	Any number of tasks
Training	You train it, per task	Already pre-trained
Scale	~30–40 terms	~350 billion parameters
New task?	Build a new model	Just ask
Fails by	Inaccurate prediction	Fluent but wrong output
Best when	Narrow, repeated, cost matters	Open-ended or varied

Task-based models do not hallucinate. They make inaccurate predictions. Do not use the two words interchangeably.

HOW AN LLM PREDICTS

“A class five student entered the class, sat on his desk and opened _____” — every answer the class gave was valid. The model picks the highest probability: **book (0.97)**. Add context about thirst after games class, and **water bottle jumps to 0.98**.

- **Context changes the prediction** — prompting works because you shift the probability distribution.
- **Output is probabilistic, not deterministic.** If your spec assumes the same input always gives the same output, you have written a bug.
- Nobody — not you, not the builders — can trace an answer to its source.

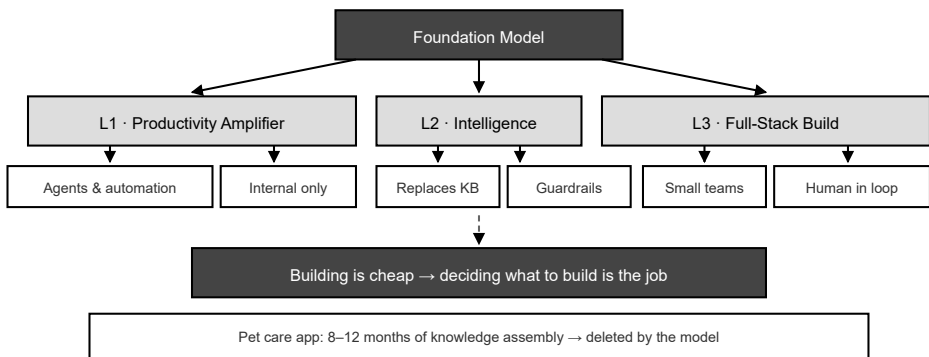
FIRM · MODEL · PLATFORM

The candidate who said “ChatGPT is a model” ended the interview. Keep three things separate:

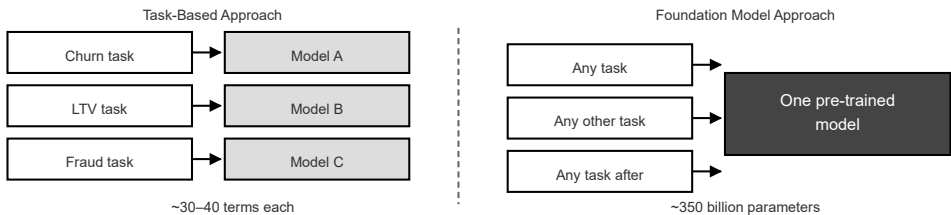
- **Firm** — the company: OpenAI, Anthropic, Google, Perplexity, xAI
- **Model** — what produces the output: GPT 5.6, Opus, Gemini 3.5
- **Platform** — the interface you reach it through: ChatGPT, Claude.ai, Claude Code

ChatGPT writing text and ChatGPT producing an image are **two different models** (GPT 5.6 for text, DALL·E 3 for images) under one front end.

VISUAL SUMMARIES



Three independent layers trace back to one shift. The moat that was a curated knowledge base is gone.



One task one model vs one model any task. The scale difference is three orders of magnitude.

OPEN SOURCE VS PROPRIETARY

- **Proprietary** (closed): cannot download; reach via API. You do not own it. OpenAI, Anthropic, Google flagships.
- **Open-source**: download weights, run on your hardware, no per-call fee. Qwen (7B, 35B), DeepSeek.
- **Hugging Face** is a platform for finding open-source models — not a model itself.

RAG VS FINE-TUNING

- **RAG** — your data stays outside the model; the model retrieves from it at answer time. Simpler, cheaper.
- **Fine-tuning** — train the foundation model further on your data, changing the model itself. More expensive. Amazon Rufus is the class example.
- Fine-tuning a proprietary model does not let you download or own it.

KEY TAKEAWAYS

- **Intelligence is becoming a commodity.** The next 5–10 years is the window where knowing how to work with it makes you disproportionately valuable.
- **Learn all three layers, not one or two.** “When you’re well versed in all three, your role is kind of secure.”
- **Building is no longer the hard part.** Deciding what to build, and making it reliable, is where the work sits.
- **Generic products are dying.** Anyone can build a nutrition app; nobody will buy one.
- **Model selection is a cost decision.** Test accuracy *and* cost together, per step, in your own domain.
- **Domain expertise still matters.** AI without knowing what to use it for is useless.

COMMON MISTAKES

VOCABULARY AND THINKING ERRORS FROM CLASS

- Calling ChatGPT a model. It is a **platform**; GPT 5.6 and DALL·E 3 are the models underneath.
- Using “hallucination” for every AI error. Task-based models make **inaccurate predictions**, not hallucinations.
- Designing as if the model is deterministic. Consistency is engineered *around* the model.
- Assuming the knowledge base is still your moat. Check if a foundation model already has it.
- Reaching for a foundation model when a small trained model would do — the cost difference at volume is enormous.
- Optimising accuracy without pricing it. Best model on every step kills your unit economics.

PRACTICE

- ☐ Build the vendor map: for each firm, list text, image, video, audio and code models plus platforms.
- ☐ Pick 10 AI features you use daily; classify each by layer with a one-line justification.
- ☐ Split one product into task-based and foundation steps; add volume estimates per step.
- ☐ Kill your own moat: try to reproduce your product’s differentiator in 15 minutes of prompting.
- ☐ Run the context experiment: same question, no context vs three sentences of context vs rephrased context. Save all three answers.

GLOSSARY

Foundation model — pre-trained on a vast corpus; any task, no retraining. ~350B parameters.

Task-based model — one task, one model. ~30–40 terms. Simpler, cheaper.

LLM — large language model; predicts the next word probabilistically.

RAG — data stays outside the model; retrieved at answer time.

Fine-tuning — training a foundation model further on your data.

Hallucination — fluent but wrong output from a foundation model.

Guardrails — constraints that keep model output safe and accurate.

Firm / Model / Platform — company, the AI engine, the interface you use.

Knowledge base — curated data a product needs; formerly the moat.

Pre-trained — already trained; no per-task retraining needed.